Multi-Objective Online Learning

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References

Chapter 3 of Aaron's lecture notes:
https://www.cis.upenn.edu/~aaroth/GamesInLearning.pdf

Group-wise Regret

Online Learning with Multiple Subsequences

For $t = 1, 2, \dots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \dots, \ell_t(N)\right) \in \mathbb{R}^N$

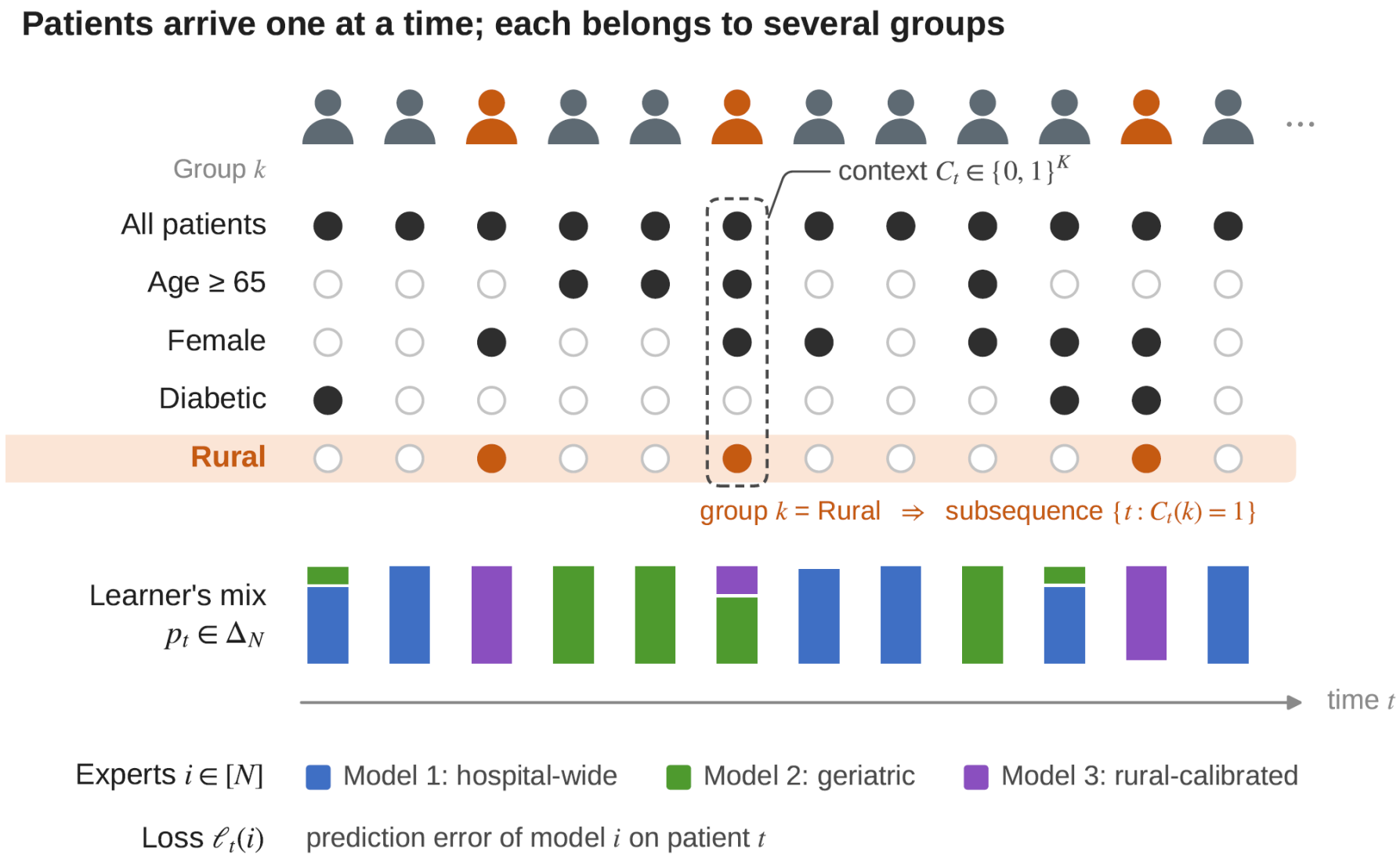
Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Online Learning with Multiple Subsequences



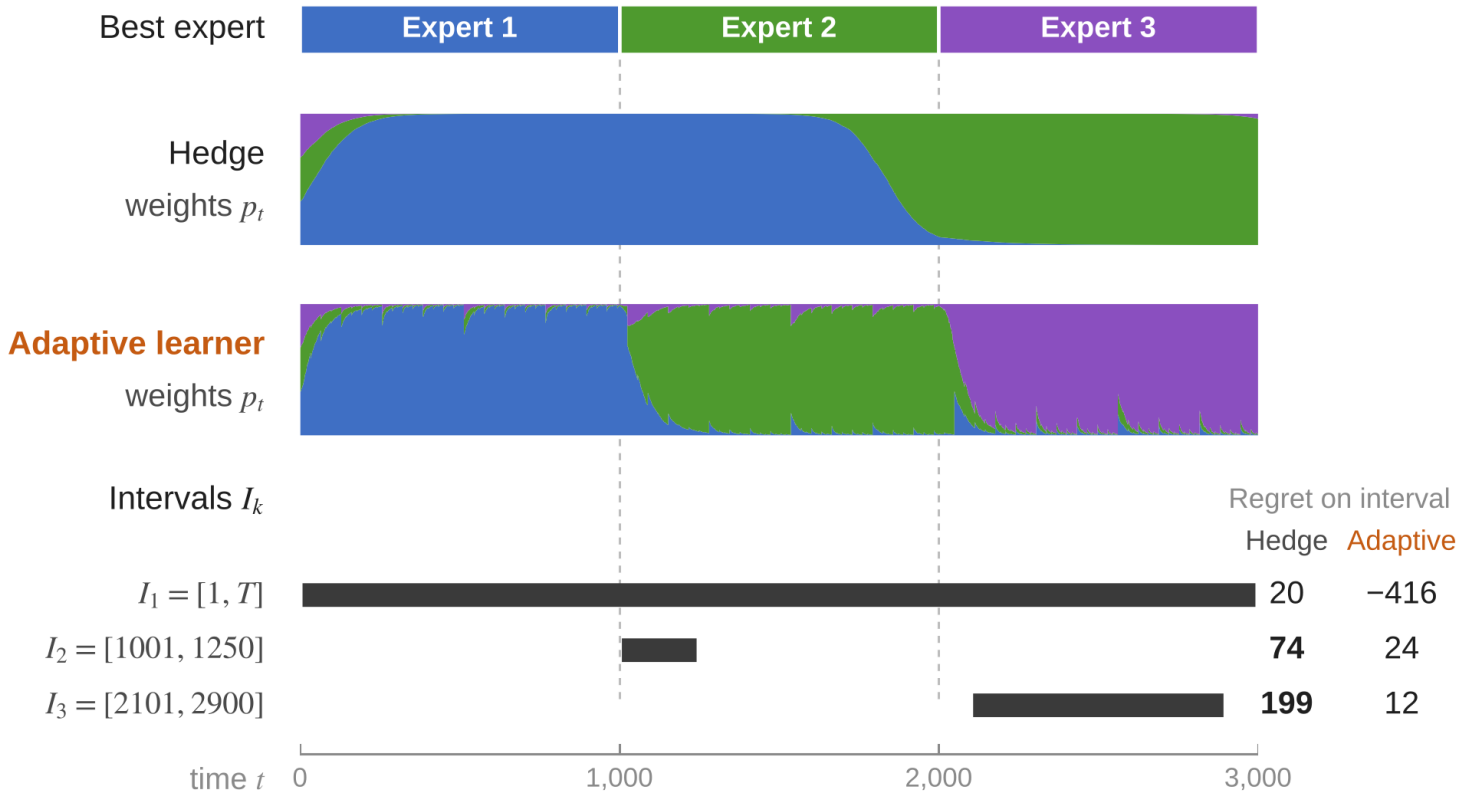
Why care about group-wise regret?

Online Learning with Multiple Subsequences

(a) The best expert changes over time; every interval is a group

Adaptive regret $= \max_{I=[s,e]\subseteq[T]} \left(\sum_{t\in I} \langle p_t, \ell_t \rangle - \min_{i\in[N]} \sum_{t\in I} \ell_t(i) \right)$

= group-wise regret with one group per interval: $C_t(k) = \mathbf{1}\{t \in I_k\}$



Is the standard no-regret guarantee sufficient?

No.

Consider loss sequence $\ell_t = (0,1)$ for $t \le \frac{T}{2}$

$\ell_t = (1,0)$ for $t \in [\frac{T}{2}, T]$

Since $L_t(1) \le L_t(2)$ on any t, if we run exponential weights on it, we have $p_t(1) \ge p_t(2)$ on any t

This makes $\sum_{t=T/2}^{T} \langle p_t, \ell_t \rangle - \sum_{t=T/2}^{T} \ell_t(2) = \Omega(T)$

Our goal

$$\max_{k\in[K]} \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{T \log(NK)}\right)$$

In fact, the following stronger guarantee is possible:

$$\forall k \in [K] \qquad \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{\left(\sum_{t=1}^{T} C_t(k)\right) \log(NK)}\right)$$

Recall: No-Regret Best-Response Dynamics

With a known $G \in \mathbb{R}^{M \times N}$

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = \underset{p \in \Delta_M}{\operatorname{argmin}}\, p^\top G q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G^\top p_t \in \mathbb{R}^N$

Boosting:	Model player	Sample player
LP:	Variable player	Constraint player

Recall: No-Regret Best-Response Dynamics

Guarantee:

$$\max_q \sum_{t=1}^{T} p_t^\top G q \le T \cdot \text{GameValue}(G) + O\left(\sqrt{T \log N}\right)$$

$\min_p \max_q p^\top G q$

By the regret guarantee of the q-player,

$$\max_q \sum_{t=1}^{T} p_t^\top G q - \sum_{t=1}^{T} p_t^\top G q_t \le O\left(\sqrt{T \log N}\right)$$

$$p_t^\top G q_t = \min_p p^\top G q_t \le \min_p \max_q p^\top G q$$

Modified Scenario

With a time-varying and unknown $G_t \in \mathbb{R}^{M \times N}$

Additional assumption:

$$\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \leq V^\star$$

$\forall q \ \min_{p} \max_{G_t} p^\top G_t q \leq V^\star$

p-player (minimizer)

For $t = 1,2, \ldots$

~~Choose $p_t = \operatorname{argmin}_{p \in \Delta_M} p^\top G q_t$~~

Choose p_t such that $p_t^\top G_t q_t \leq V^\star$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G_t^\top p_t \in \mathbb{R}^N$

No-regret robust-response dynamics

Modified Scenario

Guarantee:

$$\max_{q} \sum_{t=1}^{T} p_t^\top G_t q \le T V^\star + O\left(\sqrt{T \log N}\right)$$

By the regret guarantee of the $\{$-player.

$$\max_{\{} \sum_{t=1}^{T} p_t^\top G_t \{ - \sum_{t=1}^{T} p_t^\top G_t \{_t \le O\left(\sqrt{T \log N}\right)$$

$\{_t \le V^*$

A "trivial" instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

Verify: $\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \le 0$ ✓

$$p^\top G_t q = \sum_{i,j} p(i) q(j) G_t(i,j)$$

$$= \sum_{i,j} p(i) q(j) \left(\ell_t(i) - \ell_t(j) \right)$$

$$= \langle p, \ell_t \rangle - \langle q, \ell_t \rangle \le 0$$

Given q, just set $p = q$.

Guarantee:

$$\max_q \sum_{t=1}^{T} p_t^\top G_t q \le O\left(\sqrt{T \log N}\right)$$

$$\Rightarrow \max_q \sum_{t=1}^{T} \sum_{i,j} p_t(i) q(j) \left(\ell_t(i) - \ell_t(j) \right) \le O\left(\sqrt{T \log N}\right)$$

$$\Rightarrow \max_q \sum_{t=1}^{T} \left(\langle p_t, \ell_t \rangle - \langle q, \ell_t \rangle \right) \le O\left(\sqrt{T \log N}\right)$$

p-player's regret over loss ℓ_t

A “trivial” instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

The interaction just looks like below (somewhat trivial):

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $r_t(j) = \langle p_t, \ell_t \rangle - \ell_t(j)$

Back to "Online Learning with Multiple Subsequences"

For $t = 1, 2, \dots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \dots, \ell_t(N)\right) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Back to "Online Learning with Multiple Subsequences"

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$ $\quad i \in [N],\ j \in [N],\ k \in [K]$ (group)

Verify: $\forall q, C_t\ \exists p\ \forall \ell_t \qquad p^\top G_t q \le 0$

$W_t(j) \triangleq \sum_k q(j,k)\, C_t(k)$

$$p^T G_t q = \sum_{i,j,k} p(i)\, q(j,k)\, G_t(i,(j,k))$$

$$= \sum_{i,j,k} p(i)\, q(j,k)\, C_t(k)\big(\ell_t(i) - \ell_t(j)\big)$$

$$= \sum_i p(i)\,\ell_t(i) \sum_{j,k} q(j,k)\, C_t(k) - \sum_i p(i) \sum_j \ell_t(j) \sum_k q(j,k)\, C_t(k)$$

$$\simeq \sum_i p(i)\,\ell_t(i) \sum_j W_t(j) - \sum_i p(i) \sum_j \ell_t(j)\, W_t(j)$$

$$\sum_i p(i)\, \ell_t(i) \sum_j w_t(j) - \sum_i p(i) \sum_j \ell_t(j)\, w_t(j)$$

re-index $i \leftrightarrow j$

$$= \sum_{i,j} p(i)\, \ell_t(i)\, w_t(j) - \sum_{i,j} p(j)\, \ell_t(i)\, w_t(i)$$

$$= \sum_{i,j} \left(p(i)\, w_t(j) - p(j)\, w_t(i) \right) \ell_t(i) = 0$$

pick $p(i) = c \cdot w_t(i)$

depend on C_t

$$c\, w_t(i)\, w_t(j) - c\, w_t(j)\, w_t(i) = 0$$

Back to “Online Learning with Multiple Subsequences”

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$

Guarantee:

$$\max_{\{j\}} \sum_{t=1}^{T} p_t^T G_t \{j\} \le O(\sqrt{T \log NK})$$

$$\Rightarrow \max_{\{j\}} \sum_{t=1}^{T} \sum_{i,j,k} p_t(i)\, q(j,k)\, C_t(k)\left(\ell_t(i) - \ell_t(j)\right) \le O(\sqrt{T \log(NK)})$$

$$\Rightarrow \max_{j,k} \sum_{t=1}^{T} \sum_{i} p_t(i)\, C_t(k)\left(\ell_t(i) - \ell_t(j)\right) \le O(\sqrt{T \log(NK)})$$

$$= \max_{k} \max_{j} \sum_{t=1}^{T} C_t(k)\left(\langle p_t, \ell_t \rangle - \ell_t(j)\right)$$ ← group-wise regret

Resulting Algorithm

$p_t(i) \propto \sum_{k:\text{arm } i \text{ in } t} q_t(i,k)$

For $t = 1,2, \ldots$

Choose $p_t \in \Delta_N$ as

$$p_t(i) = \frac{\sum_k q_t(i,k) C_t(k)}{\sum_{j,k} q_t(j,k) C_t(k)} = \frac{w_t(i)}{\sum_j w_t(j)}$$

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_{NK}$ from no-regret alg

Feed no-regret alg with reward

$r_t(j,k) = C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(j))$

A No-Groupwise-Regret Algorithm

Initialize $q_t(i,k) = 1$ for all $(i,k) \in [N] \times [K]$

For $t = 1, 2, \ldots, T$:

Learner observes $C_t \in \{0,1\}^K$

Learner plays distribution

$$p_t(i) \propto \sum_{k=1}^{K} C_t(k)\, q_t(i,k)$$

Adversarial reveals $\ell_t \in \mathbb{R}^N$

Update $q_{t+1}(i,k) = q_t(i,k) \exp\left(\eta\, C_t(k)\left(\langle p_t, \ell_t\rangle - \ell_t(j)\right)\right)$

$$\max_{k\in[K]} \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t\rangle - \ell_t(i)) \le O\left(\sqrt{T \log(NK)}\right) \quad \text{with optimal choice of } \eta$$

How to Achieve A More Adaptive Performance?

$$\forall k \in [K] \qquad \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t\rangle - \ell_t(i)) \le O\left(\sqrt{\left(\sum_{t=1}^{T} C_t(k)\right)\log(NK)}\right)$$

Recall exponential weights guarantees

$$P_{t+1}(i) \propto P_t(i)\exp(-\eta \ell_t(i)) \qquad \sum_t \langle p_t, \ell_t\rangle - \sum_t \ell_t(i^*) \le \frac{\log N}{\eta} + \eta \sum_t \sum_i p_t(i)\ell_t(i)^2 \quad \forall i^*$$

$$\sum_t \langle p_t, \ell_t\rangle - \sum_t \ell_t(i^*) \le \frac{\log N}{\eta} + \eta \sum_t \ell_t(i^*)^2 \quad \forall i^*$$

$$P_{t+1}(i) \propto P_t(i)\exp\left(-\eta \ell_t(i) - \eta^2 \ell_t(i)^2\right)$$

How to Achieve A More Adaptive Performance?

$$\sum_t \langle p_t, \ell_t \rangle - \sum_t \ell_t(i^*) \le \sqrt{\left(\sum_t \ell_t(i^*)^2\right) \log(NT)} \qquad \forall i^*$$

If we use this in the update of ξ_t, then

$$\forall k, i \qquad \sum_t c_t(k)\left(\langle p_t, \ell_t \rangle - \ell_t(i)\right) \le \sqrt{\sum_t \left(c_k(t)\left(\langle p_t, \ell_t \rangle - \ell_t(i)\right)\right)^2 \log(NTK)}$$

$$\le \sqrt{\left(\sum_t c_k(t)\right) \log(NTK)}$$

Blum and Lykouris. Advancing Subgroup Fairness via Sleeping Experts. 2020.

Chen, Luo, Wei. Impossible Tuning Made Possible: A New Expert Algorithm and Its Applications. 2021.

Application: Interval Regret (a.k.a. Adaptive Regret)



$$\sum_{t=t_1}^{t_2} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=t_1}^{t_2} \ell_t(i) \lesssim O\left(\sqrt{t_2 - t_1}\right)$$

for all $1 \le t_1 < t_2 \le T$?

Let every group correspond to an interval $[t_1, t_2]$

there'll be $\frac{T(T-1)}{2}$ groups

$$\sum_t \langle p_t, \ell_t \rangle - \sum_t \min_i \ell_t(i) \lesssim \sqrt{T_1} + \cdots + \sqrt{T_S} \le \sqrt{ST}$$

$$\log\left(NT \cdot \frac{T(T-1)}{2}\right) = O(\log NT)$$



of times the best expert changes

Remark

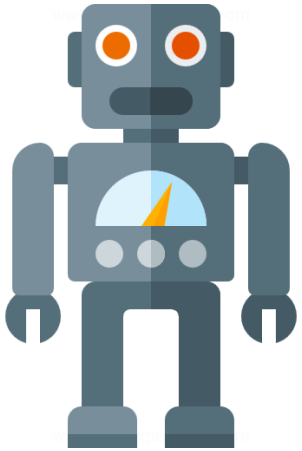
- All the derivations above for the group-wise regret actually holds with real-number $C_t(k) \in [0,1]$.



Auditor

monitors if $\exists k, i$ such that

$$\sum_t \left[c_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right) \right] \text{ is large}$$



Learner



Adversary ℓ_t

Swap Regret

Swap Regret

For $t = 1, 2, \ldots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \ldots, \ell_t(N)\right) \in \mathbb{R}^N$

Learner samples an action $i_t \sim p_t$

Adversary reveals ℓ_t to the learner

Internal Regret for Action i $= \sum_{t:\, i_t = i} \ell_t(i) - \min_{j \in [N]} \sum_{t:\, i_t = i} \ell_t(j)$

$$= \max_{j \in [N]} \sum_{t=1}^{T} C_t(i)\left(\ell_t(i) - \ell_t(j)\right)$$

$$C_t(i) = \mathbf{1}[i_t = i]$$

$\forall j$

$\sum_{t:\, i_t = i} \ell_t(i) \le \sum_{t:\, i_t = i} \ell_t(j) + o(T)$

i_t

i_t

$\le o(T)$

$P_t(i)$

Swap Regret

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \dots, \ell_t(N)\right) \in \mathbb{R}^N$

Learner samples an action $i_t \sim p_t$

Adversary reveals ℓ_t to the learner

$$\text{Swap Regret} = \sum_{i=1}^{N} [\text{Internal Regret for Action } i]$$

$$= \sum_{t=1}^{T} \ell_t(i_t) - \min_{\phi:[N]\to[N]} \sum_{t=1}^{T} \ell_t(\phi(i_t))$$

$\phi(1) = 2$
$\phi(2) = 3$
$\phi(3) = 2$

$N = 3$

Internal Regret for $i = \max_j \sum_{t=1}^{T} \mathbb{1}\{i_t = i\} \left(\ell_t(i) - \ell_t(j) \right)$

Define the $\phi(i) = \operatorname{argmin}_j \sum_{t=1}^{T} \mathbb{1}(i_t = i)\, \ell_t(j)$

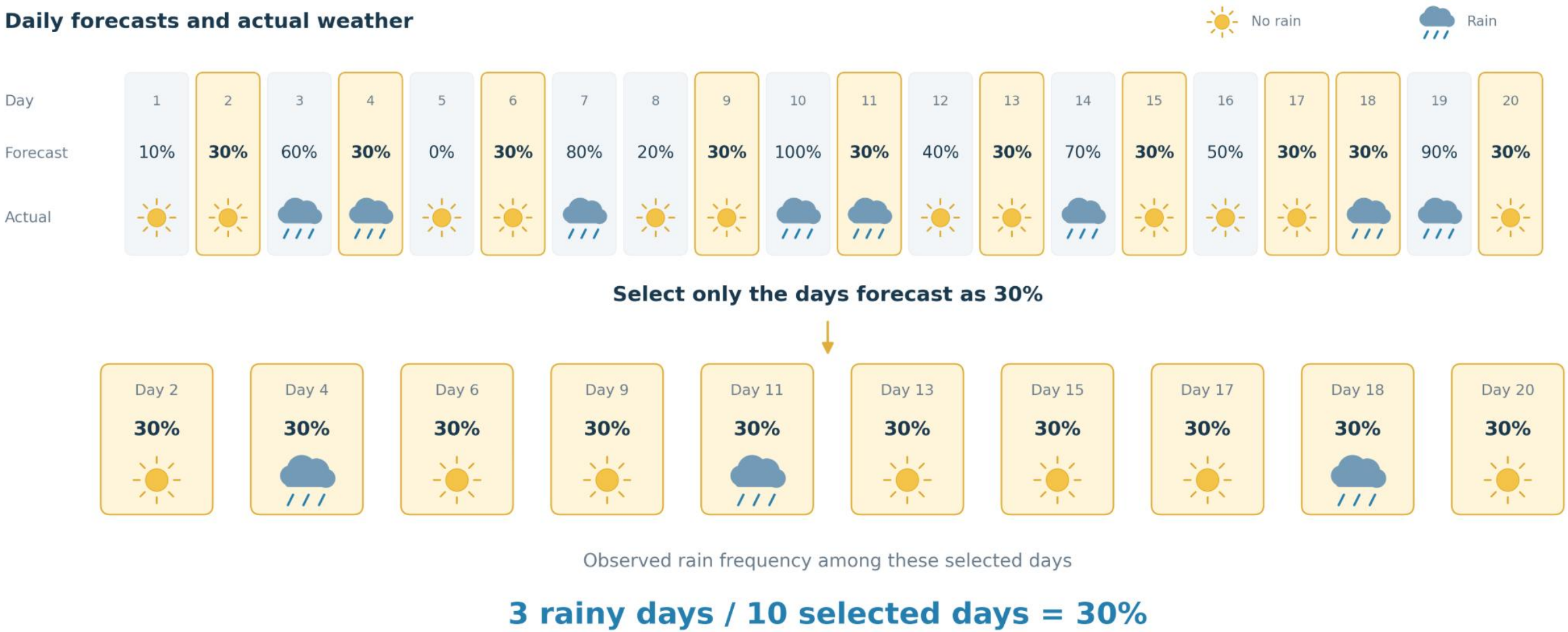
$\Rightarrow$ internal regret for i $= \sum_{t=1}^{T} \mathbb{1}\{i_t = i\} \left(\ell_t(i) - \ell_t(\phi(i)) \right)$

$\Rightarrow$ Swap regret $= \sum_{t=1}^{T} \left(\ell_t(i_t) - \ell_t(\phi(i_t)) \right)$

sum over i;

Application: Calibration

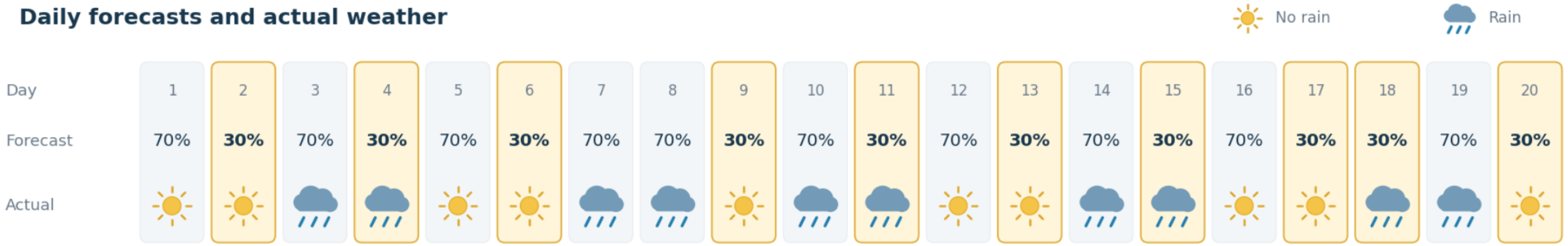
Allowed forecasts: 0%, 10%, 20%, ..., 100% chance of rain.



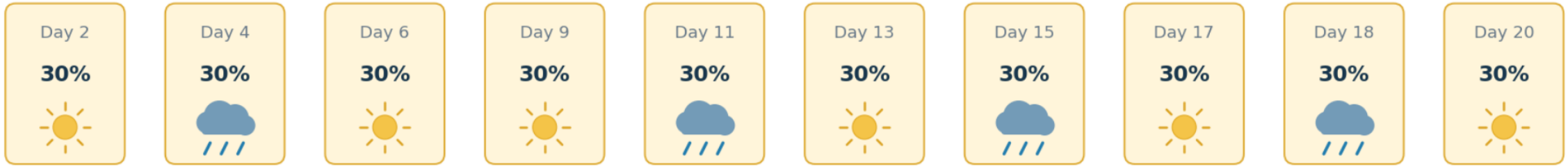
Application: Calibration

Average forecast = 50% | Overall rain frequency = 10/20 = 50%

Daily forecasts and actual weather



Select only the days forecast as 30%



Observed rain frequency among these selected days

4 rainy days / 10 selected days = 40% ≠ 30%

Total squared loss: forecaster = 5; best fixed forecast (50%) = 5

External regret = 5 − 5 = 0

No-Swap-Regret Implies Calibration

$$\ell_t(i) = (i - \alpha_t)^2 \qquad \alpha_t \in \{0, 1\}$$

i: prediction; α_t: no rain / rain

$$\text{Internal Regret for } i = \sum_{i_t = i} \ell_t(i) - \min_j \sum_{i_t = i} \ell_t(j)$$

$$\min_j \sum_{i_t = i} (j - \alpha_t)^2$$

$$\text{The minimizer } j \approx \frac{\sum_{i_t = i} \alpha_t}{\sum_{i_t = i} 1}$$

Recall the Framework

With a time-varying and unknown $G_t \in \mathbb{R}^{M \times N}$

Additional assumption:

$\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \leq V^\star$ 0

p-player (minimizer)

For $t = 1,2, \ldots$

Choose p_t such that $p_t^\top G_t q_t \leq V^\star$ 0

q-player (maximizer)

For $t = 1,2, \ldots$ (action, group)

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G_t^\top p_t \in \mathbb{R}^N$

No-regret robust-response dynamics

A Game for No-Swap Regret Learning

$G_t \in \mathbb{R}^{N \times N^2}$

$G_t(i,(j,k)) = 1[i=k](\ell_t(i) - \ell_t(j))$

Guarantee:

$$\max_q \left\{ \sum_{t=1}^{T} p_t^T G_t q \right\} \le o(T) \quad (V^* = 0)$$

$$\Rightarrow \max_{j,k} \sum_{t=1}^{T} (p_t^T G_t)_{j,k} \le o(T) \Rightarrow \max_{j,k} \sum_{t=1}^{T} \sum_i p_t(i) \mathbb{1}\{i=k\} (\ell_t(i) - \ell_t(j)) \le o(T)$$

$$\Rightarrow \max_{j,k} \sum_{t=1}^{T} p_t(k) (\ell_t(k) - \ell_t(j)) \le o(T)$$

$$\Rightarrow \max_{j,k} \sum_{t=1}^{T} \mathbb{1}\{i_t=k\} (\ell_t(k) - \ell_t(j)) \le o(T) + o(T)$$

A Game for No-Swap Regret Learning

$G_t \in \mathbb{R}^{N \times N^2}$

$G_t(i,(j,k)) = 1[i=k](\ell_t(i) - \ell_t(j))$

$$\sum_k \left(p(k) \sum_j g(j,k) - \sum_j p(j) g(k,j) \right) \ell_t(k)$$

Find p such that

$$p(k) \sum_j g(j,k) = \sum_j p(j) g(k,j)$$

Verify: $\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \le 0$

$$p^T G_t g = \sum_{i,j,k} p(i) g(j,k) \mathbf{1}\{i=k\} (\ell_t(i) - \ell_t(j))$$

$$= \sum_{j,k} p(k) g(j,k) (\ell_t(k) - \ell_t(j))$$

$$= \sum_{j,k} p(k) g(j,k) \ell_t(k) - \sum_{j,k} p(k) g(j,k) \ell_t(j)$$

$$= \sum_{j,k} p(k) g(j,k) \ell_t(k) - \sum_{j,k} p(j) g(k,j) \ell_t(k)$$

Let $Q(j,k) = q(j,k) + \mathbf{1}[j=k]\left(1 - \sum_{i=1}^{N} q(i,k)\right)$

Then $Q(j,k)$ is an $N \times N$ transition matrix because every entry is non-negative and

$$\sum_{k=1}^{N} Q(j,k) = \sum_{k=1}^{N} q(j,k) + 1 - \sum_{k=1}^{N} q(i,k) = 1.$$

Let $p \in \Delta_N$ be the stationary distribution under Q, i.e., $Qp = p$.

$Qp = p$ implies

$$\begin{aligned} p(j) &= \sum_{k=1}^{N} Q(j,k)p(k) = \sum_{k=1}^{N} \left(q(j,k) + \mathbf{1}[j=k]\left(1 - \sum_{i=1}^{N} q(i,k)\right)\right) p(k) \\ &= \sum_{k=1}^{N} q(j,k)p(k) + \left(1 - \sum_{i=1}^{N} q(i,j)\right) p(j) \end{aligned}$$

and thus $\left(\sum_{k=1}^{N} q(k,j)\right) p(j) = \sum_{k=1}^{N} q(j,k)p(k)$, which is what we want in the previous slide (with j,k indices swapped).

A No-Swap-Regret Algorithm

Initialize $q_t(i,k) = 1$ for all $(i,k) \in [N] \times [K]$

For $t = 1, 2, \ldots, T$:

Define $Q_t(i,k) = q(i,k) + 1[i = k](1 - \sum_{j=1}^{N} q(j,k))$

Let p_t be such that $p_t = Q_t p_t$

Adversarial reveals $\ell_t \in \mathbb{R}^N$

Update $q_{t+1}(i,k) = q_t(i,k) \exp\left(\eta\, p_t(k)\left(\ell_t(k) - \ell_t(i)\right)\right)$